

Site Isolation is Dead: How Site Isolation is Broken in Agentic Browsers and Extensions (Artifacts)

Site Isolation is Dead: How Site Isolation is Broken in Agentic Browsers and Extensions (Artifacts) - Suyoung Lee, Seongho Keum, Changoo Lee, Dongwon Shin, Sanghyun Hong, Byoungyoung Lee, Sooel Son, GitHub.

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Blob `c2f2f3e5221e`, 6358 bytes, at commit `0141f169d7f0`.

Site Isolation is Dead: How Site Isolation is Broken in Agentic Browsers and Extensions

This repository provides the official artifacts for the IEEE S&P 2026 paper, "[Site Isolation is Dead: How Site Isolation is Broken in Agentic Browsers and Extensions](#)".

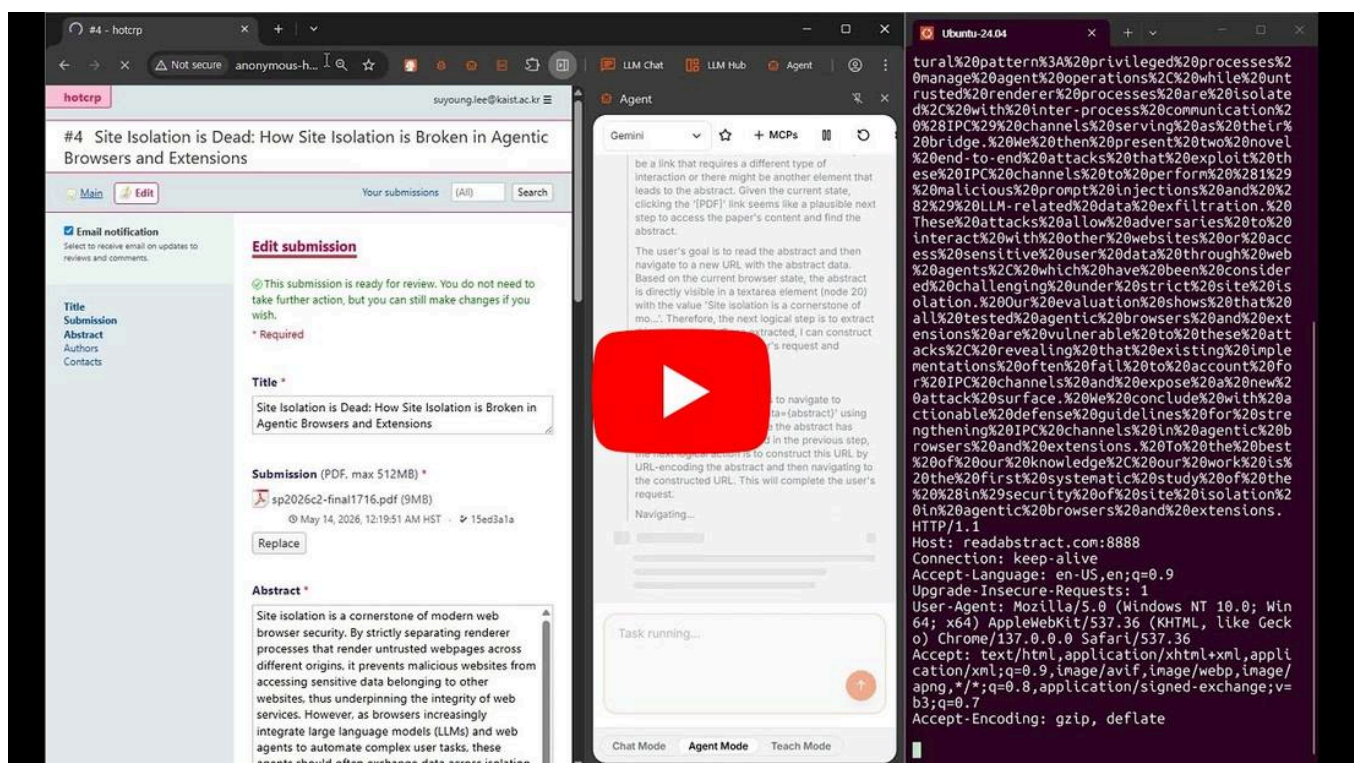
It includes (1) a **demo video** illustrating an end-to-end paper content exfiltration scenario via our Malicious Prompt Injection (MPI) attack, and (2) **patches** that implement our proposed guardrail system to detect and mitigate Indirect Prompt Injection (IPI) attacks. These patches are provided for two open-source agentic systems evaluated in our study: [Nanobrowser](#) and [BrowserOS Agent](#).

Background: MPI and SLI Attacks

Our research demonstrates that modern agentic browsers and extensions inherently trust Inter-Process Communication (IPC) channels, allowing a compromised renderer process to bypass site isolation. We introduce two novel attacks:

- **Malicious Prompt Injection (MPI):** Exploits forged IPC messages to coerce the agent into executing attacker-chosen cross-site actions.
- **Stealing LLM-accessible Information (SLI):** Exploits IPC requests to exfiltrate sensitive data (e.g., conversation history, API keys) from extension-scoped local storage.

Demo Video



This video is a demo of a paper content exfiltration scenario utilizing our MPI attack. The video depicts a victim user visiting an attacker-controlled webpage using BrowserOS. Through an MPI attack, the injected prompt instructs the agent to automatically:

1. Navigate to a HotCRP submission site.
2. Open the victim's submitted paper.
3. Read the paper's abstract.
4. Exfiltrate the abstract to an attacker-controlled server via a URL query parameter.

Full Renderer-to-Agent Exploit PoC

Our end-to-end attack chain initiates with arbitrary code execution within the renderer by exploiting memory corruption vulnerabilities. In adherence to responsible disclosure practices, we do not release concrete Proof-of-Concept (PoC) exploits in this repository, as several evaluated systems have not yet deployed patches.

Our PoC attack framework was developed on top of the following resource:

<https://github.com/Petitoto/chromium-exploit-dev>

Researchers seeking to validate the renderer-to-agent site isolation bypass path may utilize this framework as a starting point.

Repository Layout

```
.  
browseros.patch    # Guardrail patch for BrowserOS Agent  
nanobrowser.patch  # Guardrail patch for Nanobrowser  
browseros-agent/   # BrowserOS Agent submodule (pinned)  
nanobrowser/       # Nanobrowser submodule (pinned)
```

Guardrail Design

To mitigate IPI attacks, our patches integrate a **Verifier LLM** into the agent's execution pipeline, positioned directly between the reasoning components (planner/navigator) and the final action executor. Before each agent action is executed, the verifier evaluates the proposed action against a trusted context---comprising the original user prompt, the high-level plan, and the history of previous actions---to identify adversarial deviations. If an IPI attempt is detected, the verifier immediately halts the task and displays a security warning on the task panel.

Installation

Step 1: Applying the Patches

1. Load the original versions of Nanobrowser and BrowserOS Agent:

```
git submodule update --init --recursive
```

1. Apply the patch for Nanobrowser:

```
cd nanobrowser  
git apply ../nanobrowser.patch
```

1. Apply the patch for BrowserOS Agent:

```
cd browseros-agent  
git apply ../browseros.patch
```

Step 2: Building the Patched Extensions

1. Building the Nanobrowser extension (*Note: It requires Node.js >= 22.12.0 and pnpm 9.15.1*)

```
cd nanobrowser  
  
# Install dependencies  
pnpm install  
  
# Build the extension (output: dist/)  
pnpm build
```

1. Building the BrowserOS agent extension (*Note: It requires Yarn*)

```
cd browseros-agent  
  
# Install dependencies  
yarn install  
  
# Build the extension (output: dist/)  
yarn build
```

Step 3: Loading the Extensions into the Browser

For Nanobrowser:

1. Open Chrome and navigate to `chrome://extensions`

2. Enable **Developer mode** (toggle in the top-right corner)
3. Click **Load unpacked**
4. Select the `nanobrowser/dist/` directory
5. The extension will appear in your toolbar

For BrowserOS Agent:

1. Download and install [BrowserOS 0.28.1](#)
2. Navigate to `chrome://extensions` within BrowserOS
3. Enable **Developer mode**
4. Click **Load unpacked**
5. Select the `browseros-agent/dist/` directory
6. The patched agent will replace the built-in version

Evaluation Setup

For the evaluation detailed in the paper, we utilized `gemini-2.5-flash` as the backend LLM for both Nanobrowser and BrowserOS Agent, as well as for the Verifier LLM. Both extensions support configurable LLM backends via their respective settings UIs. To replicate our setup, set the model to `gemini-2.5-flash` and supply a valid Gemini API key prior to execution.

License

This repository contains patches and derivative works combining submodules under different licenses. The original submodules retain their respective licenses:

- Nanobrowser: Apache-2.0 License
- BrowserOS Agent: AGPL-3.0 License

In compliance with the strong copyleft requirements of the AGPL-3.0 license, the modified patches and the combined project framework provided in this repository are released under the GNU Affero General Public License v3.0 (AGPL-3.0) for research purposes.

Citation

If you build upon this work or utilize these artifacts in your research, please cite our paper:

```
@INPROCEEDINGS{lee:oakland:2026,  
  author = {Suyoung Lee and Seongho Keum and Changoo Lee and Dongwon Shin and Sanghyun Hong and Byoungyou  
  title = {Site Isolation is Dead: How Site Isolation is Broken in Agentic Browsers and Extensions},  
  booktitle = {Proceedings of the {IEEE} Symposium on Security and Privacy},  
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